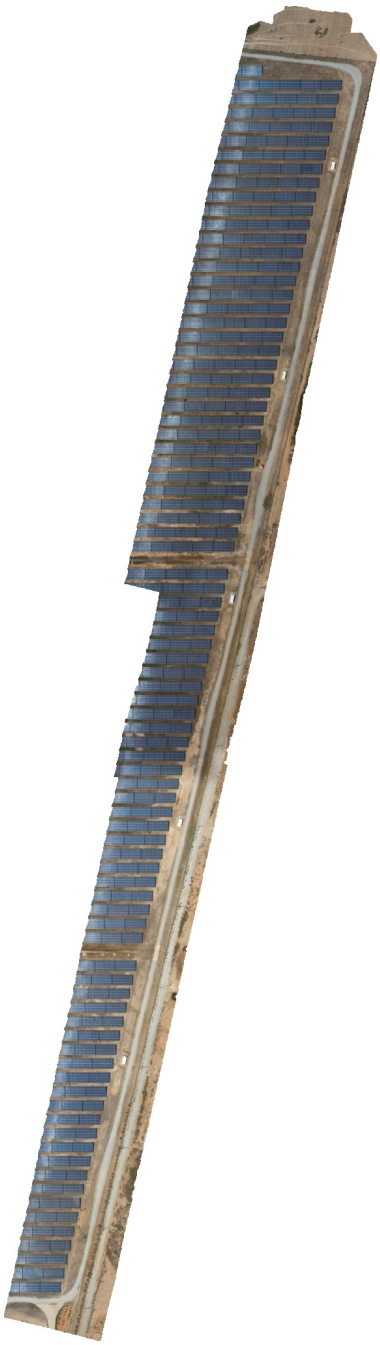
Agisoft Metashape

Processing Report

08 September 2026



Survey Data

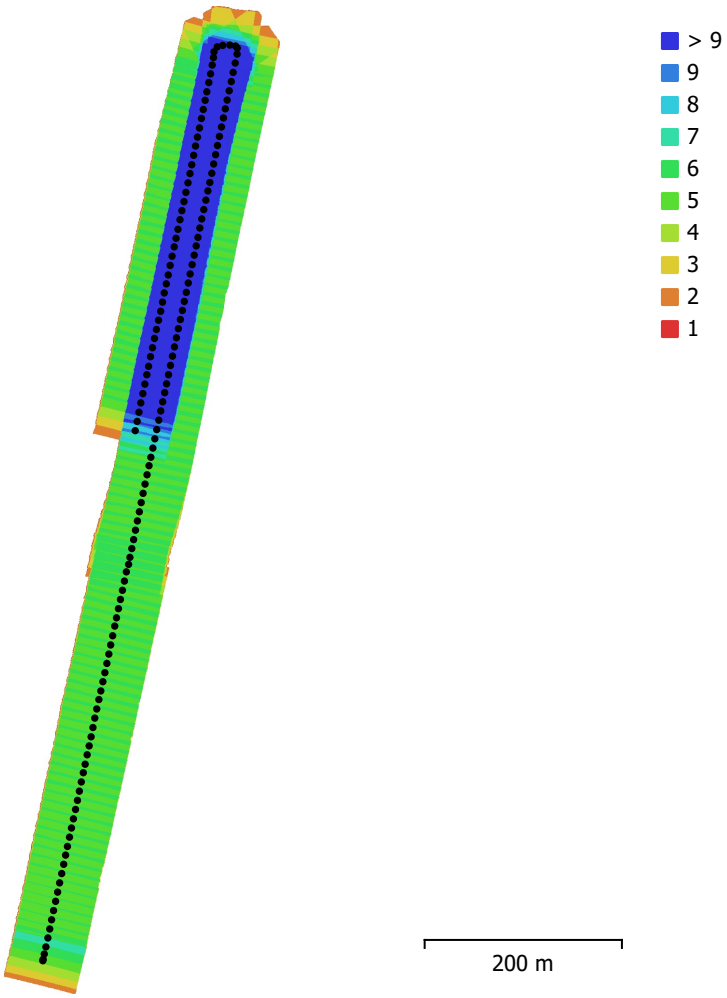


Fig. 1. Camera locations and image overlap.

Number of images:	150	Camera stations:	150
Flying altitude:	106 m	Tie points:	86,320
Ground resolution:	9.20 mm/pix	Projections:	427,506
Coverage area:	0.0884 km²	Reprojection error:	0.183 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
ZenmuseP1 (50mm)	8192 x 5460	50 mm	4.39 x 4.39 µm	No

Table 1. Cameras.

Camera Calibration

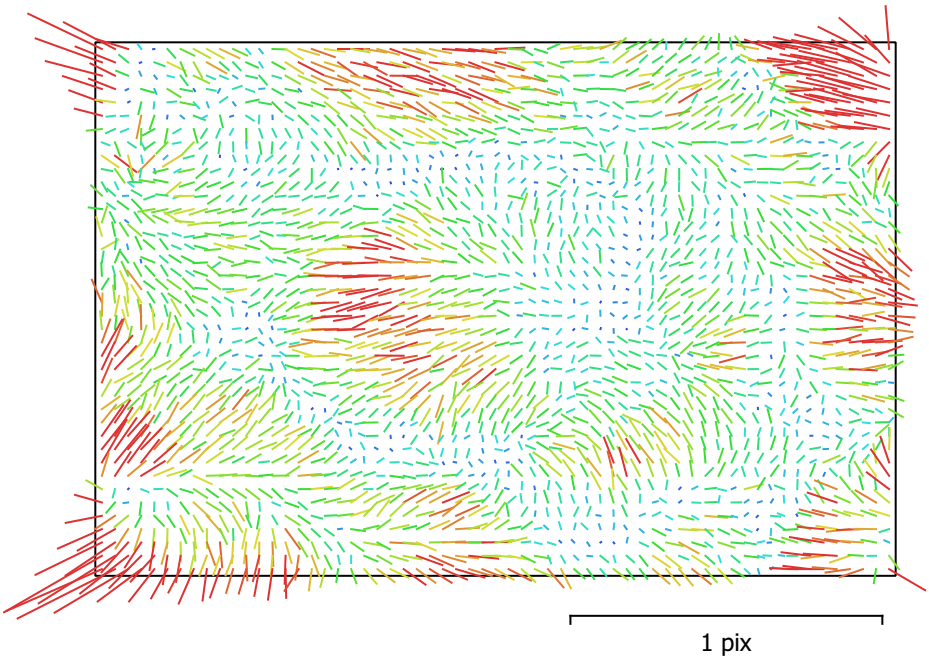


Fig. 2. Image residuals for ZenmuseP1 (50mm).

ZenmuseP1 (50mm)

150 images

Type	Resolution	Focal Length	Pixel Size
Frame	**8192 x 5460**	**50 mm**	**4.39 x 4.39 µm**

	Value	Error	Cx	Cy	B1	B2	K1	K2	K3	P1	P2
F	**11376.9**										
Cx	**2.51257**	0.077	1.00	-0.07	-0.04	-0.04	-0.06	-0.00	0.02	-0.27	-0.03
Cy	**1.55559**	0.063		1.00	0.01	0.06	-0.04	-0.01	0.01	-0.00	0.03
B1	**1.7299**	0.016			1.00	-0.06	0.12	-0.02	0.00	0.22	0.06
B2	**-0.326825**	0.014				1.00	0.06	0.00	-0.01	0.02	0.20
K1	**-0.0154265**	8.1e-05					1.00	-0.36	0.22	0.57	0.61
K2	**-0.280243**	0.0004						1.00	-0.97	0.01	0.03
K3	**-0.605942**	0.0014							1.00	-0.10	-0.11
P1	**-0.00037378**	2e-06								1.00	0.37
P2	**-4.48419e-05**	2.1e-06									1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

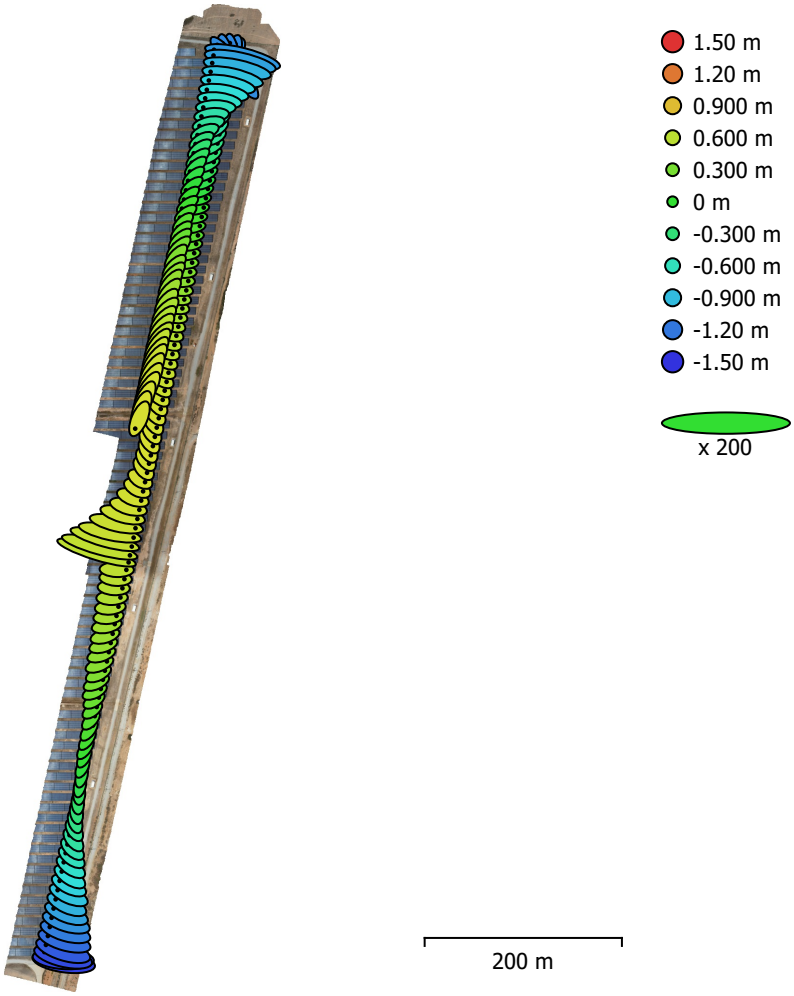


Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (cm)	Y error (cm)	Z error (cm)	XY error (cm)	Total error (cm)
10.2355	6.39722	63.1756	12.0702	64.3184

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

Digital Elevation Model

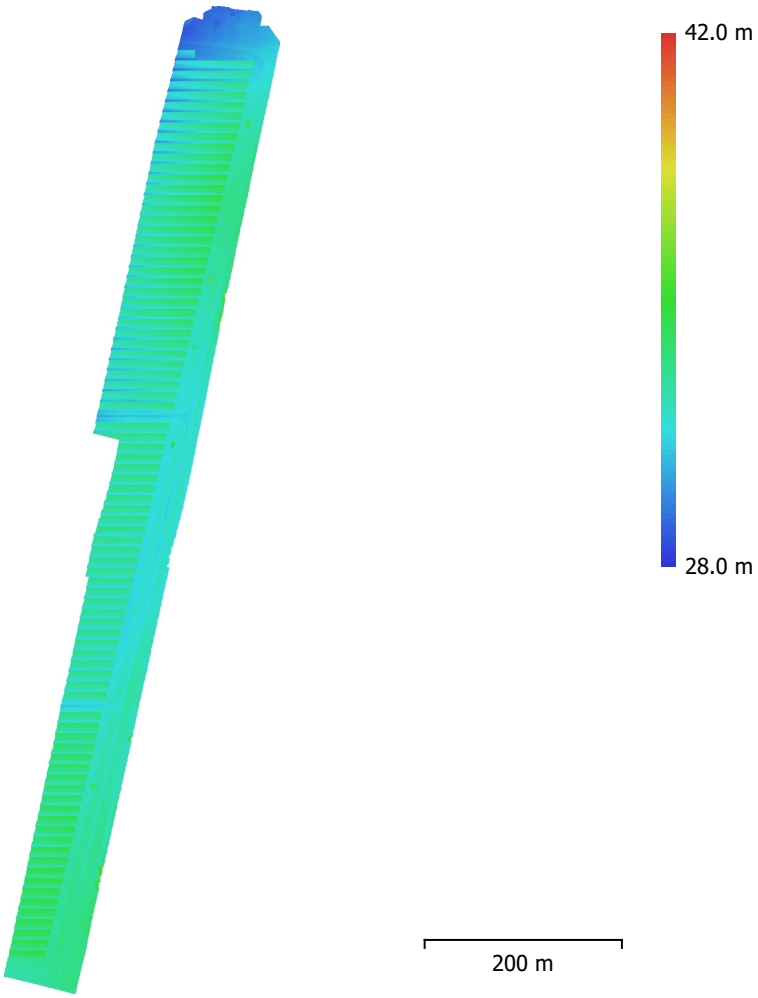


Fig. 4. Reconstructed digital elevation model.

Resolution:	1.86 cm/pix
Point density:	0.290 points/cm²

Processing Parameters

General	
Images	150
Aligned images	150
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll
Tie Points	
Points	86,320 of 132,102
RMS reprojection error	0.0806823 (0.183348 pix)
Max reprojection error	0.377951 (0.942221 pix)
Mean key point size	2.48768 pix
Point colors	3 bands, uint8
Key points	493.24 MB
Average tie point multiplicity	4.81919
Alignment parameters	
Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	Yes
Matching time	4 minutes 22 seconds
Matching memory usage	395.45 MB
Alignment time	3 minutes 41 seconds
Alignment memory usage	357.86 MB
Optimization parameters	
Parameters	f, cx, cy, k1, k2, p1, p2
Adaptive camera model fitting	Yes
Exclude corners	No
Optimization time	2 seconds
Date created	2026:08:12 10:58:24
Software version	2.3.1.22580
File size	507.76 MB
Depth Maps	
Count	150
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	31 minutes 37 seconds
Memory usage	5.50 GB
Date created	2026:08:12 11:45:01
Software version	2.3.1.22580
File size	2.17 GB
Point Cloud	
Points	283,944,399
Coordinate precision	9.42 m
Point attributes	

Color	3 bands, uint8
Normal	
Point classes	
Created (never classified)	283,944,399
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	31 minutes 37 seconds
Memory usage	5.50 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	32 minutes 33 seconds
Memory usage	5.11 GB
Date created	2026:08:12 12:17:42
Software version	2.3.1.22580
File size	3.73 GB
DEM	
Size	15,043 x 53,843
Resolution	1.86 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	31 minutes 37 seconds
Memory usage	5.50 GB
Reconstruction parameters	
Source data	Depth maps
Interpolation	Enabled
Processing time	1 hours 55 minutes
Memory usage	5.84 GB
Date created	2026:08:12 14:14:10
Software version	2.3.1.22580
File size	1001.95 MB
Orthomosaic	
Size	30,086 x 107,686
Resolution	9.29 mm/pix
Coordinate system	WGS 84 (EPSG::4326)
Colors	3 bands, uint8
Orthophotos	3.00 GB
Reconstruction parameters	
Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	16 minutes 11 seconds
Memory usage	4.05 GB
Date created	2026:08:12 14:21:18
Software version	2.3.1.22580
File size	4.45 GB
System	
Software name	Agisoft Metashape Professional
Software version	2.3.0 build 21954
OS	Windows 64 bit
RAM	15.78 GB

CPU	11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz
GPU(s)	NVIDIA GeForce RTX 3050 Laptop GPU